Mission 03: Squadron Winograd Operational Documentation

Multi-Head Attention, RoPE Vector Rotation & Grouped-Query KV-Cache Optimization

Starship Operational Unit: USSE Noosphere Bridge Crew

Course: Conversational AI — Winter Semester 2026/27

Module: Week 3 — Modern Transformer Architectures and Architectural Divergence

Galactic Waypoint: Pleiades Open Cluster (SEC-03-M45, 444 ly from Sol — Seven Sisters)

AURA Subsystem Repair: Subsystem 03: Distributed Multi-Head Attention Mesh & KV-Cache Transceiver (+7.1% $\to$ 21.3% total core health)

IDE & Tools Standard: Visual Studio Code + Antigravity CLI (`agy`) | PyTorch 2.4+

Squadron: Squadron Winograd

Repository: Public Squadron Winograd GitHub Repository (Folder `mission-03/`)

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1. Squad Roster & Mission Sign-Offs

Station Role	Officer Name	Assigned Module / Deliverable	Status	Authorization Hash
Systems Architect	Muhammad Qasim Siddiqui	Part 1: `01_MultiHead_Attention_Core.py`	**PASS / VERIFIED**	`SIG-ARCH-QAS-3101`
Communications & Protocol Officer	Muhammad Usman Maratib	Part 2: `02_RoPE_Positional_Engine.py`	**PASS / VERIFIED**	`SIG-PROT-USM-8443`
Data & Payload Specialist	Hassaan Ahmad Khawaja	Part 3: `03_GQA_and_KVCache_Benchmark.py`	**PASS / VERIFIED**	`SIG-DATA-HAS-3319`

Security & Ethics Sentinel	Muhammad Qasim Siddiqui	Part 4: `04_Memory_Hierarchy_and_Safety_Audit.md`	**PASS / VERIFIED**	`SIG-SECU-QAS-7192`

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2. System Architecture & Attention Mesh Topology

```
flowchart TD
  subgraph AstrogationInput ["Pleiades M45 Optical Resonance Stream"]
    direction TB
    PhotonicStream["Multi-Spectral Reflection Stream<br/>Context: N = 32,768 Tokens"]
  end

  subgraph AttentionMesh ["AURA Subsystem 03: Transformer Core"]
    direction TB
    LinearProj["Q, K, V Linear Projections<br/>From Scratch (Wq, Wk, Wv)"]
    RoPE_Rot["Rotary Position Embedding<br/>Complex 2D Rotation (m * theta_i)"]
    KVCache_Buffer["Stateful KV-Cache Buffer<br/>O(1) Autoregressive Decoding"]
    GQA_Broadcasting["GQA Head Expansion (32:8)<br/>repeat_interleave(4x)"]
    ScaledDotProduct["Scaled Dot-Product Attention<br/>Variance Preserved: 1.0 / sqrt(d_k)"]

    LinearProj --> RoPE_Rot --> KVCache_Buffer --> GQA_Broadcasting --> ScaledDotProduct
  end

  subgraph HardwareExecution ["Memory & Hardware Defense"]
    direction TB
    SRAM_Tiling["FlashAttention Block Tiling<br/>Online Softmax (Zero HBM Write)"]
    PagedBuffer["PagedAttention Block Memory<br/>Zero Fragmentation Defense"]
    TerminalOutput["Astrogation Solution Locked<br/>Ship Readiness: 21.3%"]

    ScaledDotProduct --> SRAM_Tiling --> PagedBuffer --> TerminalOutput
  end

  PhotonicStream --> LinearProj

  classDef input fill:#0e7490,stroke:#22d3ee,stroke-width:2px,color:#ffffff;
  classDef core fill:#0f172a,stroke:#38bdf8,stroke-width:2px,color:#e2e8f0;
  classDef defense fill:#064e3b,stroke:#10b981,stroke-width:2px,color:#a7f3d0;
  class PhotonicStream input;
  class LinearProj,RoPE_Rot,KVCache_Buffer,GQA_Broadcasting,ScaledDotProduct core;
  class SRAM_Tiling,PagedBuffer,TerminalOutput defense;
```

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3. Station Metrics, Traces & Key Findings

3.1 Systems Architect (Multi-Head Self-Attention from Scratch)

- **Station Lead:** Muhammad Qasim Siddiqui
- **Deliverable:** `01_MultiHead_Attention_Core.py`
- **Linear Projections:** Explicit $W_Q, W_K, W_V, W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ without library abstractions.
- **Causal Masking:** Strict lower-triangular causal attention mask ($-\infty$ above diagonal).
- **Variance Preservation:** Validated empirically that scaling dot products by $1/\sqrt{d_k}$ normalizes input variance from 63.81 down to 0.9971 (~ 1.0000).

3.2 Protocol Officer (RoPE Positional Engine)

- **Station Lead:** Muhammad Usman Maratib
- **Deliverable:** `02_RoPE_Positional_Engine.py`
- **Rotary Complex Embedding:** Planar vector rotation using precomputed frequencies $\text{cis}(m \cdot \theta_i)$.
- **Translation Invariance:** Validated $\langle \text{RoPE}(q, m), \text{RoPE}(k, n) \rangle = \langle \text{RoPE}(q, m+\delta), \text{RoPE}(k, n+\delta) \rangle$ with absolute error $< 10^{-5}$ (0.0000019).
- **Distance Decay Curve:** Mapped natural decay against ALiBi linear penalty curves.

3.3 Data Specialist (GQA & KV-Cache Benchmarking)

- **Station Lead:** Hassaan Ahmad Khawaja
- **Deliverable:** `03_GQA_and_KVCache_Benchmark.py`
- **Head Grouping:** Configurable $H_Q:H_{KV}$ (32:8) with `repeat_interleave` expansion.
- **Stateful Caching:** In-memory `KVCache` retaining past tokens across autoregressive steps.
- **VRAM Benchmark (32 Layers, FP16):**
- Context $N = 32,768$: MHA = 16.38 GB, GQA = 4.10 GB (75% reduction), MQA = 512 MB.
- Memory Bandwidth at 50 tok/s: MHA = 800 GB/s vs. GQA = 200 GB/s.

3.4 Security Sentinel (FlashAttention & DoS Audit)

- **Station Lead:** Muhammad Qasim Siddiqui
- **Deliverable:** `04_Memory_Hierarchy_and_Safety_Audit.md`
- **FlashAttention IO Derivation:** Proved HBM reads/writes drop from $O(N^2)$ to $O(N^2 d / M)$ via SRAM block tiling and online softmax.
- **Denial-of-Service Scenario:** Simulated catastrophic memory fragmentation under unconstrained multi-turn dialogue, presenting PagedAttention and Multi-Head Latent Attention (MLA) mitigations.

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4. Full Test-Suite Execution Output

```
============================= test session starts =============================
platform win32 -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: E:\Study\Conversational AI History, Applications, Future Seminar
plugins: anyio-4.14.2
collecting ... collected 6 items

Winograd Squad/mission-03/tests/test_ps03.py::test_mha_tensor_shapes PASSED [ 16%]
Winograd Squad/mission-03/tests/test_ps03.py::test_mha_causal_masking PASSED [ 33%]
Winograd Squad/mission-03/tests/test_ps03.py::test_mha_variance_preservation PASSED [ 50%]
Winograd Squad/mission-03/tests/test_ps03.py::test_rope_tensor_shape_preservation PASSED [ 66%]
Winograd Squad/mission-03/tests/test_ps03.py::test_rope_relative_translation_invariance PASSED [ 83%]
Winograd Squad/mission-03/tests/test_ps03.py::test_gqa_kv_cache_stateful_decoding PASSED [100%]

============================== 6 passed in 1.42s ==============================
```

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5. Squadron Dialectic Reflection

Reconstructing the core Transformer attention pipeline in Sector SEC-03-M45 highlights the essential evolution from theoretical attention to hardware-aware execution:

1. **Relative Geometric Invariance:**

Standard absolute positional embeddings fail when extrapolating beyond training context windows. RoPE's formulation—embedding positional information as complex rotations in orthogonal 2D planes—proves that relative token distance is the natural geometric language of sequential cognition.

2. **Overcoming the Memory Wall:**

Quadratic attention ($O(N^2)$) and full MHA caching fail not because ALUs lack compute, but because HBM bus bandwidth cannot feed memory fast enough. Grouped-Query Attention and FlashAttention SRAM tiling bridge this chasm, allowing our starship AI to maintain long-context coherence without exceeding physical hardware limits.